

MATH120 Discrete Mathematics

Tutorial 1

References

- Topic 1: Logic
- Sections 2.1, 2.2, 2.4 of Epp (4th or 5th Ed.)

Work on the following problems in groups during the tutorial in week 2.

1. [B] Construct a truth table for each of the following compound statements, where p and q denote primitive statements.

(a) $(p \wedge (p \rightarrow q)) \rightarrow q$

(b) $q \leftrightarrow (\neg p \vee \neg q)$

2. [M] Determine all truth values assignments, if any, for the primitive statements p, q, r, s, t that make the following compound statement false:

$$((p \wedge q) \wedge r) \rightarrow (s \vee t).$$

3. [M] Let p, q, r be primitive statements. Use truth tables to verify the following logical equivalence:

$$((p \vee q) \rightarrow r) \Leftrightarrow ((p \rightarrow r) \wedge (q \rightarrow r)).$$

- $\Rightarrow$
4. [M] Determine whether the following statements are logically equivalent. In each case, construct a truth table, and explain your answer.

(a) $p \vee (p \wedge q)$ and p .

(b) $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$.

5. [B] Use truth tables to establish which of these statements are tautologies, and which are contradictions.

(a) $(p \wedge q) \vee (\neg p \vee (p \wedge \neg q))$

(b) $(p \wedge \neg q) \wedge (\neg p \vee q)$

(c) $(\neg p \vee q) \vee (p \wedge \neg q)$

6. [B] Write each of these statements in symbolic form, and determine whether they are logically equivalent. Include a truth table and explanation in your answer.

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- $\implies$ 7. [M] Write a negation of the following statements.

(a) If n is prime, then n is odd or n is 2.

(b) If n is divisible by 6, then n is divisible by 2 and n is divisible by 3.

8. [M] Use $\leftrightarrow$ to convert this logical equivalence to a tautology. Then use a truth table to verify it is a tautology.

$$p \rightarrow (q \vee r) \iff (p \wedge \neg q) \rightarrow r$$

9. [M] Consider the NAND statement form: $p \bar{\wedge} q$, which is equivalent to $\neg(p \wedge q)$.

For each of the following, construct a truth table, and identify which single logical connective could replace the NAND form.

(a) $\neg p \bar{\wedge} \neg q$

(b) $p \bar{\wedge} \neg q$